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Quantum Technologies in the UK

Risks and Opportunities for National Security

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Quantum physics is the branch of science that describes the behaviour of the world of the small; it defines the properties of atoms, molecules, electrons and photons. Quantum science has been explored for decades, but practical applications have only emerged in recent years. Through its National Quantum Technologies Programme and other investments, the UK is one of the leading countries driving innovation in quantum computing, security and sensing. These technologies promise transformative impact across multiple sectors, from healthcare to finance, defence, energy and national security.

As with all technologies, quantum devices pose threats to national security: quantum computers can be applied to attack encryption standards; quantum security could keep communications by nefarious groups secure; and quantum sensors may be used to track military operations that would otherwise be considered covert.

Ensuring that the UK retains its competitive position in developing quantum technologies and establishing a robust supply chain demands significant investment in specialised resources and expertise, both technical and operational.

An Introduction to Quantum Technology

The defining capabilities of quantum devices are derived from the fundamental properties of quantum mechanics that they embrace. These include:

- Quantum confinement: the wave-like, probabilistic nature of 'particles' such as electrons and photons and how restricting the motion of these influences their behaviour. This property also relates to a phenomenon known as quantum tunnelling, which can allow particles to pass through barriers forbidden in classical physics.
- Superposition: the fact that quantum systems can exist in multiple states simultaneously.
- Entanglement: the fact that quantum systems can behave as if they are joined or in a shared state, even when far apart.

These properties underpin the capabilities of quantum devices to compute, securely communicate and sense in ways that traditional technologies cannot.

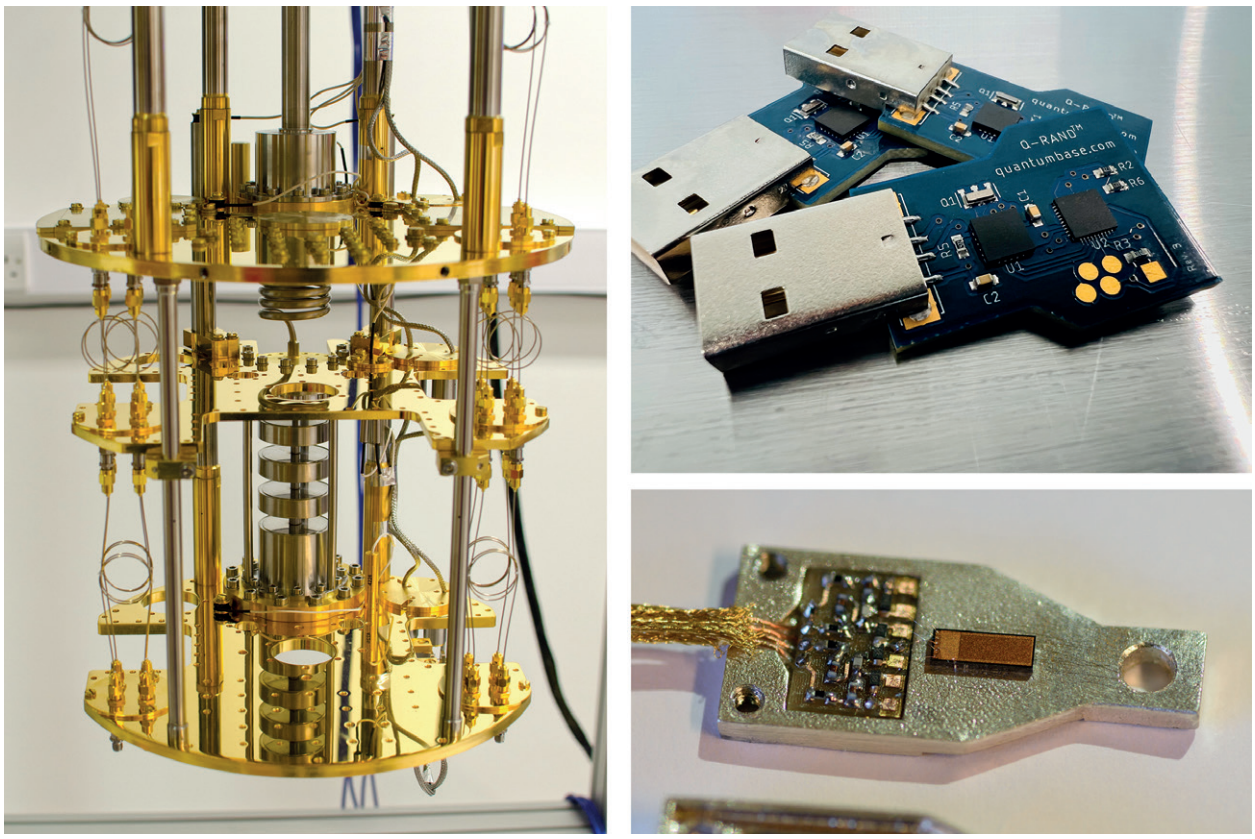
The UK's National Quantum Technologies Programme kickstarted significant momentum in this domain by funding research, fostering industrial partnerships, and setting up dedicated quantum technology hubs. This approach aims to help the UK remain a key player in quantum research, patents, publications and industrial investment.

The US and China dominate quantum technology development due to their massive research budgets and strong private sector involvement. China's patent filings have been particularly prolific in recent years. Commercially, large US tech firms (such as IBM, Google, Microsoft, Honeywell and Intel) are pushing forward with quantum computing platforms, while Chinese corporations (such as Alibaba and Baidu) are accelerating their national efforts. In Europe, Germany, the Netherlands and France are front-runners, driven by substantial government funding. The UK has demonstrated a strong commitment to quantum technologies through substantial investment and a long-term vision, having invested more than £1 billion to date, with an aim at accelerating research, innovation and commercialisation to establish the UK as a global leader in the field. Private investment in the sector in the UK is rapidly approaching £1 billion pewr annum.

Understanding the 'Tech' in Quantum Technology

Quantum technologies are emerging. Half a century ago, it was impossible to predict where the advent of the digital revolution would lead us – and we are in a similar position today when evaluating the future potential of the quantum technology revolution. It has branched into three primary subfields, each with distinct operational principles, implementation requirements, and applications. The subsections below give an overview of quantum computing, sensing, and security, emphasising the technical details that highlight how each technology works and why it is strategically relevant.

Figure 1: Quantum computing (left), sensing (bottom right) and security devices (top right).



Source: Courtesy of Quantum Technology Centre, Lancaster University.

Quantum Computing

Basic Principles

Computers traditionally represent and store information in binary digits, 0s and 1s (bits). Quantum computers are fundamentally different, normally processing information in superpositions of a 0 and 1 state encoded on quantum bits (qubits, two-level quantum systems); measurements and reading out information from qubits collapses superpositions into binary states that can represent the results of calculations. In quantum computation, entanglement between qubits can be used to scale the complexity of the problems that can be tackled.

Implementation

Many different systems for realising qubits have been proposed. The leading contenders today are:

- Superconductors: cooled to temperatures close to absolute zero, superconducting circuits allow qubits to be manipulated via precisely timed microwave pulses. Companies like IBM, Google and Oxford Quantum Circuits are at the forefront of this approach – IBM and Google reported significant advancements recently, defining the boundaries of qubit performance and count. Figure 2 shows a photograph of IBM's Quantum System One.
- Ions: electromagnetic fields can trap and manipulate charged atoms (ions). Lasers and microwaves are used to manipulate quantum states encoded on these ions. IonQ, Quantinuum and Alpine Quantum Technologies are pioneers here.
- Photons: particles of light– photons– can be transmitted and processed through optical circuits. Photonic systems can operate at room temperature and allow for modular scaling. Xanadu, Quandela and ORCA Computing are pioneering companies here.

The lack of a leading technology for implementing qubits for computation indicates that material advancement is still required in this field, as no qubit system is ideal.

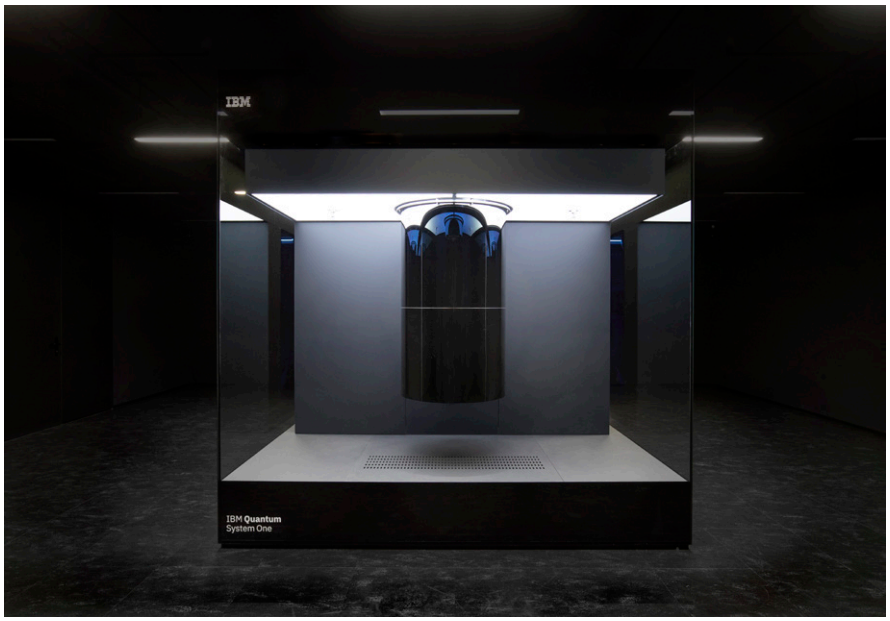
Quantum simulators are an implementation used to model complex physical systems that are subtly different from general-purpose quantum computers. They use carefully controlled quantum systems to mimic the behaviour of other quantum or classical systems, which are then used to provide insights into phenomena that would otherwise be hard to model. Unlike universal quantum computers, these devices target specific problems, which makes them more

resource-efficient. Through this direct mirroring of the quantum dynamics of the system under study, they offer powerful routes of investigation.

Applications

- Cryptanalysis: large-scale quantum computers have long been heralded as having the potential to attack current public-key cryptography schemes such as RSA, elliptic curve cryptography and discrete log-based schemes (such as Diffie–Hellman). Thus, much of today's encrypted communications are under threat.
- Chemical and materials simulation: quantum devices can accurately model molecular interactions, enabling breakthroughs in drug discovery, energy storage and weapon design.
- Logistics and optimisation: quantum algorithms can tackle complex optimisation problems relevant to defence logistics, supply chain management and strategic planning.

Figure 2: Quantum computers: IBM's Quantum System One



Source: IBM Research / Flickr / CC BY 2.0.

Quantum Sensing and Metrology

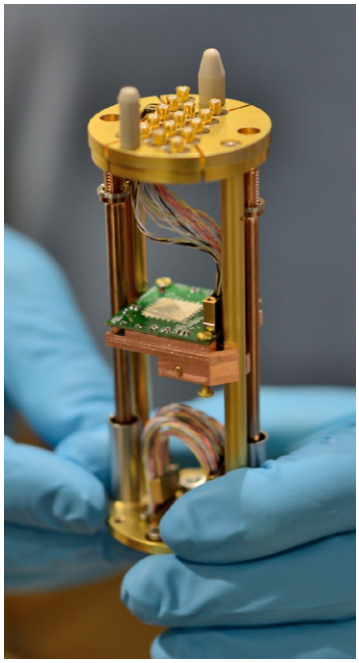
Basic Principles

Quantum sensors use quantum states, often entangled or in superpositions, to measure magnetic, electric or gravitational fields, electromagnetic radiation, or time. They target performance metrics, such as signal-to-noise ratios, that are better than conventional devices and, in some cases, better than the theoretical maximum classical performance that is possible. Quantum states are susceptible to small changes, and quantum sensors often exploit this to detect minute variations that classical sensors cannot detect.

Implementation and Applications

- Atomic clocks use atomic properties to precisely define a period (tick) to monitor time passing. They can be highly stable reference devices crucial for GPS, maintain synchronisation in high-speed telecommunication systems, and underpin transactions in financial ecosystems.
- Gravimeters detect tiny fluctuations in gravitational fields with subterranean and subsea mapping applications.
- Magnetometers measure extremely weak magnetic fields. They are used in biomedical imaging, submarine detection and the identification of concealed metallic objects.
- Photon detectors are essential for many applications, from secure communications to gas and molecular sensing. Developments in photon detectors have driven applications in quantum imaging, which aims to achieve resolutions and signal-to-noise ratios beyond classical limits. Ghost imaging uses correlations between entangled photons to reconstruct an image, even when the detector does not directly view the object. Advancements here could be important in biomedical imaging and military contexts, offering advantages in surveillance, reconnaissance and target identification under low-light or adverse conditions.

Figure 3: A graphene superconducting quantum interference device (SQUID).



Source: Dr Michael Thompson, Lancaster University's Quantum Technology Centre.

Quantum Security

Basic Principles

A common misconception is that quantum technology can only be used to decrypt data. The field of quantum security targets the opposite: applying quantum physics to keep information secure and provide solutions with provable metrics. The first example of this, proposed in the 1970s (although unplublished until 1983), is known as quantum money. The scheme stores randomised quantum states in banknotes. Unknown superposition states cannot be cloned or measured without errors, thus preventing counterfeiting. While not currently practical, this idea led to the invention of quantum key distribution, which uses a similar idea to prove that communications are free from eavesdropping.

Another simple, elegant application emerged from the realisation that the probabilistic nature of quantum mechanics could be used to generate random numbers. The need for randomness in applications is widespread, from gambling to scheduling and simulations. However, it is critical in the security field: if the generation of cryptographic keys is predictable, then security is compromised. Many companies have developed and are developing quantum random number generators, ranging from scalable solutions for large-scale applications (such as the Internet of Things (IoT) to ones that do not require trust in the supplier.

Applications

Quantum Key Distribution (QKD)

Quantum states, encoded on photons, can be used to distribute encryption keys. Eavesdropping alters these states, alerting parties to the intrusion.

Multiple pilot QKD links and networks have been implemented in the UK. Different protocols have been developed alongside methods to combine classical and quantum communications on the same communication links.

However, current implementations are still slow and inefficient. Current research aims to tackle practical challenges. Entangled photons, for example, can support more advanced secure communication protocols that lay the groundwork for a future quantum internet, where quantum states are transmitted and processed across distributed networks. Toshiba Europe, based in Cambridge, is the leading developer.

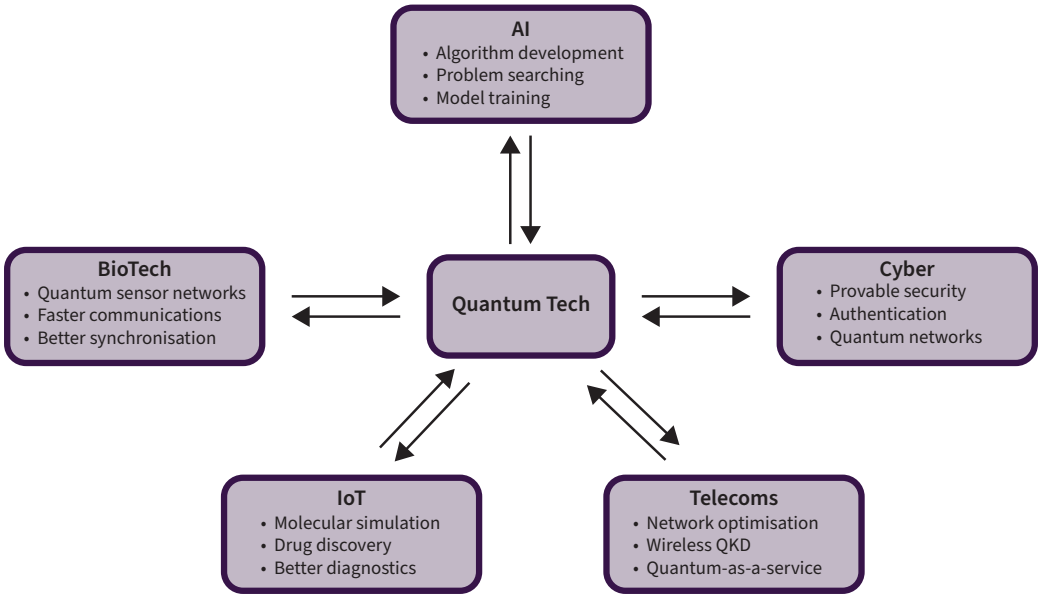
Other Applications

This field also encompasses applications for authentication, proof of provenance and anti-counterfeiting. Physically unclonable functions link imperfections, often introduced in a manufacturing process, to the identity of an object. They can harness quantum effects to generate 'fingerprints' sensitive to atomic imperfections. Quantum Base is pioneering this technology in the UK.

Intersections with Other Technologies

Quantum technologies are not being developed in isolation. They intersect with other nascent fields, forming synergies to realise novel capabilities.

Figure 4: Intersection between Quantum Technologies and Others



Source: The author.

- **AI**: today, quantum computers have limited resources and capabilities that are hamstrung by errors. AI may help identify problems that can be efficiently tackled with imperfect quantum computation and develop the algorithms employed. In the future, quantum computing may help accelerate the generation of machine learning models.
- **Cyber Security**: as quantum computing threatens cryptographic standards, post-quantum cryptography (PQC) has emerged to encrypt data using algorithms that cannot be practically attacked by either a classical supercomputer or a future 'all-powerful' quantum computer. Some lattice-based schemes appear to be resistant to any computational attacks, and the US National Institute of Standards and Technology recently published FIPS 203, 'Module-Lattice-Based Key-Encapsulation Mechanism Standard', a draft set of standards incorporating PQC algorithms. The institute also aims to select and recommend quantum-resistant key-encapsulation mechanisms

for standardisation soon. Key challenges for PQC are efficient implementation, encouraging (or mandating) widespread adoption and, crucially, gaining confidence that the accepted standards are secure. There is a long track record of standardised encryption schemes being compromised.

- **5G/6G Telecommunications**: quantum computing algorithms can be applied to network optimisation, accelerating the allocation of spectrum and reducing latency in ultra-dense network environments. Wireless quantum communication has been demonstrated, primarily using laser links and telescopes, but there is potential for implementation using radiowaves/microwaves.
- **IoT**: quantum sensors may enable hyper-accurate geolocation or environmental monitoring when integrated into IoT networks. For instance, quantum clocks synchronised across an IoT network could improve the timing accuracy in smart city applications.
- **Biotechnology**: quantum computing's potential to compute complex molecular interactions far more efficiently than classical methods stands to accelerate drug discovery by simulating how potential compounds will bind to targets. This may enable personalised medicine. Quantum sensors could detect subtle variations in biological signals and biomarkers, enhancing disease detection. Quantum devices may be applied to secure valuable genomic data and patient records and maintain confidentiality in the data-driven biotech industry.

The confluence of quantum technology with AI, cyber security, telecommunications, IoT and biotech underscores its broader transformative potential.

Relevance of Quantum Technologies for National Security

National security concerns around quantum technologies revolve around opportunity and risk. Quantum computing, sensing and security boost intelligence capabilities and strategic defence operations, while adversarial use compromises cyber security and covert operations.

- **Cryptography and Secure Communication**: perhaps the most immediate national security concern is the potential for quantum computers to break widely used public-key encryption protocols. An attack, often called a 'harvest now, decrypt later' attack, could be employed today where valuable encrypted data is stored in anticipation of future decryption aided by more advanced technologies. In response, the UK's National Cyber Security Centre supports initiatives in PQC that use algorithms that are believed to resist quantum attacks. However, such resistance is difficult to prove, and researchers are trying to develop quantum algorithms targeting lattice problems.
- **Intelligence, Surveillance and Reconnaissance (ISR)**: sensors underpin ISR, and the potential for quantum sensors to enhance ISR capabilities is important, especially in quantum imaging. UK-based start-up QLM Technology is developing quantum light detection and ranging systems for high-resolution gas sensing, which has potential implications for military intelligence.
- **Supply Chain Dependencies**: quantum hardware implementation has specialised requirements. Quantum computers and sensors often require cryogenic refrigeration, advanced production facilities are needed for creating superconducting qubits, ultra-high vacuum chambers are used for ion traps, and advanced photonic components underpin most quantum communication devices. These requirements create dependencies on specific materials and manufacturing processes. Ensuring a robust domestic supply chain could be critical. Strengthening manufacturing capacity would mitigate apparent risks associated with supply chain disruptions.

The UK hosts a vibrant quantum ecosystem with numerous leading quantum technology companies and a collaborative environment for them to thrive, as provided by the UK's National Quantum Technologies Programme and National Quantum Computing Centre. 771 UK-based quantum-related companies were

studied in Beauhurst's Quantum Ecosystem Mapping report, published in March 2025. Prominent examples in quantum computing include: Nu Quantum; ORCA Computing; Oxford Quantum Circuits; Quantinuum; Quantum Motion; Riverlane; and Universal Quantum. Example in quantum security include: Aegiq; Arquit; KETS Quantum Security; PQShield; Quantum Base; Quantum Dice; and Toshiba Europe. Examples in quantum sensors include: Cerca Magnetics; MoniRail; Peratech; QLM Technology; and Quantum Detectors. Key enabling technologies include Oxford Instruments and M Squared Lasers.

Future Development and Horizon Scanning

The trajectory of quantum technologies is evolving rapidly, driven by scientific breakthroughs and commercial investment. Predicting the shape and size of the iceberg, the tip of which we have only recently discovered, is difficult. Below are some short-, medium- and long-term predictions.

Five-Year Horizon

Noisy intermediate-scale quantum devices, featuring hundreds of qubits, have demonstrated a quantum advantage. They will soon target useful, though modest, problems. These devices will not break classical cryptography and will target specific tasks such as drug discovery and materials design.

Early-stage commercial solutions will continue to emerge. Quantum-as-a-service models, offered via cloud platforms, will expand. These will enable businesses and government agencies to experiment with quantum algorithms without owning hardware.

In quantum-ready cyber security, PQC standards will be increasingly adopted, and expanded pilots of quantum communication links in government and corporate networks will be implemented.

10-Year Horizon

Quantum computers with 1-10k qubits and improved error correction will likely enable meaningful cryptanalysis of specific classical encryption schemes and broader logistics, as well as AI and cryptography applications.

Small-scale quantum networks with entanglement-based secure communication could emerge in metropolitan areas or specialised government facilities. This evolution aligns with ongoing efforts to build quantum repeaters facilitating long-distance entanglement distribution.

As the quantum supply chain matures in the UK, it should also become more robust. Local manufacturing of key components like cryogenic systems, advanced photonic chips, and specialised electronics will reduce reliance on imports.

20-Year Horizon

Advances in error correction, scaling and algorithm flexibility will enable large-scale, fault-tolerant quantum computers that can tackle classically intractable problems. Large-scale quantum computers could range from complex modelling to codebreaking.

Quantum sensors may become ubiqitous, integrated into everyday technology such as atomic clocks in mobile devices, enabling unprecedented accuracy in navigation and communication bandwidths.

Widespread adoption of global quantum protocols and standards will likely reshape global cyber security practices. Nations leading these standards may gain disproportionate influence, further elevating the strategic dimension of quantum technology.

McKinsey & Company estimates the potential quantum technology market size in 2040 to be over $100 billion.

Long-term investment in quantum research, talent development and enabling infrastructure is a strategic imperative for the UK. Quantum technologies will be pivotal in shaping the nation's future technological and economic leadership, delivering transformative capabilities that drive innovation, productivity and growth. In an era of intensifying global competition, the UK's national security and strategic preparedness will increasingly hinge on its ability to harness these technologies and remain ahead of global competitors.

About the Author

Rob Young is Chief Scientist and co-founder of Quantum Base Limited (QBL), as well as a Professor in the Quantum Technology Centre at Lancaster University in the UK. He has more than 20 years of experience in the field, having co-authored more than 100 scientific publications and patents. His career in quantum science began with an award-winning PhD from the University of Cambridge, which focused on novel devices producing entangled photos. He went on to work in Tyndall National Institute in Ireland, developing a protocol to distribute quantum information to homes. In 2009, he moved to Lancaster and from there, started QBL a few years later. QBL is focused on applying simple quantum technologies to acute problems in security, having produced more than half a billion Q-ID devices with its partners.

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